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Association between Glucagon-Like Peptide 1 (GLP-1) Receptor Agonists Exposure and Intraocular Pressure Change

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Conflict of Interest

Dr. Baxter reports consulting fees from Topcon and equipment support from Topcon and Optomed outside the submitted work. Dr. Cui reports a provisional patent application related to the topic of this study (publication number: US20220193201A1). SH, WH, and BGC: None. Dr. Weinreb reports consulting fees from Abbvie, Alcon,Balance, Editas, Eyenovia, iSTAR Medical, Nicox, and Topcon outside the submitted work. He also has stock options from Balance, Eyenovia, Iantrek, Implandata,and Toku outside the submitted work and intellectual property licensed from UCSD to Toromedes outside the submitted work.

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Précis

In the paired analysis, glucagon-like peptide 1 receptor agonists were associated with intraocular pressure change. This decrease was statistically significant in the semaglutide and liraglutide subgroups and was more prominent in glaucomatous eyes.

Abstract (max: 350)

Objective

This study aims to provide data on the effects of glucagon-like peptide 1 receptor (GLP-1R) agonists on intraocular pressure (IOP).

Design

Retrospective cohort study.

Subjects, Participants, and/or Controls

1247 glaucoma surgery and treatment naïve eyes of 626 patients who were initiated on GLP-1R agonists compared to 1083 glaucoma surgery and treatment naïve eyes of 547 patients who were initiated on other oral antidiabetics.

Methods, Intervention, or Testing

The University of California Health Data Warehouse was queried for patients exposed to GLP-1R agonists or other oral antidiabetics. Index date was defined as the date of first exposure to the medication. Eyes with at least one pre-exposure and one post-exposure tonometry record within 365 days of the index date were included in the analysis. Clinical and laboratory data elements were extracted from the database. Eyes were censored from the analysis upon exposure to glaucoma hypotensive medication or glaucoma surgery. ΔIOP was analyzed using a paired t-test. Regression analysis was conducted using generalized estimating equations (GEE) accounting for inter-eye correlation. Sensitivity analyses were performed to assess the robustness of the findings.

Main Outcome Measures

Primary outcome measure was ΔIOP after exposure to the medication.

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Results

The median age of all included subjects was 66.2 years [IQR=18.3]; 607 (51.7%) were female, and 667 (56.9%) were Caucasian. Median pre-exposure IOP, HbA1c, and BMI were 15.2 mmHg [IQR=3.8], 7.5 [IQR=2.4], and 29.8 [IQR=9.4], respectively. 776 individuals (66.1%) had diabetes, with the median number of active oral antidiabetics being 1.0 [IQR=1.0], and 441 (37.5%) being insulin users. Several pre-exposure characteristics significantly differed between the GLP-1R agonist and the control group. The mean ΔIOP was -0.4±2.8 mmHg (paired t-test $p<0.001$) and -0.2±3.3 mmHg (paired t-test $p = 0.297$) in the GLP-1R agonist and other antidiabetics groups, respectively. Pre-exposure IOP was the only independent predictor of ΔIOP in multivariable GEE. Sensitivity analyses yielded similar results.

Conclusions

Although GLP-1R agonists were significantly associated with a decrease in IOP in the paired analysis, they were not associated with ΔIOP in multivariable GEE. Moreover, the difference between the ΔIOP in the two groups was small. Future prospective studies following a standardized dose and delivery method may provide further insights.

Keywords: Glucagon-Like Peptide-1 Receptor agonists, Intraocular Pressure, Hypoglycemic Agents, Glaucoma, Semaglutide, Liraglutide

Introduction

The pathophysiology of glaucoma is complex, multifaceted, and not fully understood.[1–3] It has been hypothesized that elevated intraocular pressure (IOP) and impaired retinal blood flow are initiating factors in glaucomatous optic nerve damage.[3] The onset of retinal neurodegeneration in glaucoma may also be influenced by cumulative oxidative stress in conjunction with chronic mechanical and vascular stress.[4] Oxidative stress is closely linked to cellular senescence, mitochondrial dysfunction, excitotoxicity, and neuroinflammation, all contributing factors to glaucoma progression.[5–7] Similar age-related mechanisms could also contribute to trabecular meshwork dysfunction and elevate IOP in open-angle glaucoma.[8–12]

The incretin hormone glucagon-like peptide-1 (GLP-1) plays a pivotal role in glucose homeostasis.[13] GLP-1 receptor (GLP-1R) agonists are long-acting GLP-1 mimetics first prescribed for controlling blood glucose levels in type 2 diabetes, a disease that has been proposed as a risk factor for glaucoma.[14] GLP-1R activation has been linked

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to other beneficial effects, including weight loss, protection against nonfatal stroke and cardiovascular death.[15] Multiple existing studies suggest a potential link between GLP-1R activation and protection against neuroinflammation in age-related neurodegenerative processes. Specifically, GLP-1R agonists initiate a signaling cascade that inhibits the production of proinflammatory cytokines and monocyte activation, contributing to the slowing of neurodegeneration.[16–18] Preclinical studies identified similar mechanisms of neuronal death in glaucoma, whereby GLP-1R agonists ameliorate retinal inflammation and astrogliosis to rescue retinal ganglion cells. [19–28] Clinically, retrospective case-control studies have shown up to a 44% decrease in the hazard of a new glaucoma diagnosis in diabetic patients exposed to GLP-1R agonists compared to unexposed controls.[29,30]

Interestingly, a preclinical study investigating the effects of topically administered GLP-1R agonists demonstrated evidence of IOP reduction in a mouse model of ocular hypertension.[31] In this study, we assessed the association between exposure to these medications and ΔIOP in humans by analyzing real-world data from the University of California Health Data Warehouse (UCHDW) as an initial step towards investigating whether existing preclinical findings of IOP modification can be translated into the clinic.

Methods

Data source

UCHDW was used for this study. This de-identified database uses the vendor-neutral and open-source Observational Medical Outcomes Partnership (OMOP) Common Data Model (CDM).[32] It collates electronic health records (EHR) and medical claim information of 8.7 million patients seen across all University of California (UC) Health sites since 2012.[33] This study was approved by the University of California San Diego Institutional Review Board, which also granted a waiver of informed consent.

Glucagon-Like Peptide-1 Receptor Agonist Group

Eyes with at least one IOP record in patients who were initiated on a new GLP-1R agonist (i.e., exenatide, albiglutide, dulaglutide, lixisenatide, semaglutide, or liraglutide) were queried from the UCHDW (Figure 1). The index date was set to the date of the first GLP-1R agonist exposure, as recorded in the “drug_exposure_start_date” field of the OMOP “drug_exposure” table.[32] Patients were required to be 18 years or older, have been exposed to a

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GLP-1R agonist, and have undergone at least one tonometry measurement both pre- and post-exposure within 12 months from the index date. Conversely, individuals under 18 years of age, without tonometry records within 12 months from the index date, who underwent IOP lowering procedures or surgeries before the index date, or who have a history of exposure to glaucoma hypotensive medications before the index date were excluded from the study.

Other Antidiabetics Group

The other antidiabetics group was comprised of patients not exposed to GLP-1R agonists, who initiated a new oral antidiabetic in one of the following subclasses: dipeptidyl peptidase 4 (DPP-4) inhibitors, sodium-glucose co-transporter 2 (SGLT2) inhibitors, sulfonamides, sulfonylureas, or thiazolidinediones. Eyes in this group were also required to meet all inclusion and exclusion criteria outlined above. This was chosen as the control group to allow consistency in methodology, e.g. analysis of IOP measurements around an index date linked to medication exposure. Eyes exposed to both GLP-1R agonists and other antidiabetics were included only in the GLP-1R agonist group. Only the first exposure to the medications was included in the analysis. Eyes were censored from the analysis if they were initiated on a hypotensive glaucoma medication or received an IOP lowering procedure or surgery. After applying inclusion and exclusion criteria, the subset of data available for this study included all IOP records in the database from November 4, 2017, to November 4, 2023. See Supplemental Table 1 for codes used in this study.

Outcomes and covariates

The primary outcome measure was the difference between pre-exposure and post-exposure IOP (ΔIOP). Pre-exposure IOP was defined as the mean of the three most recent readings within 365 days before exposure. Post-exposure IOP was the mean of all readings within 365 days after exposure. The secondary outcome measure was the difference between pre- and post-exposure IOP variance. Demographic covariates assessed at the time of the index date were age, race, and gender. Ocular health factors included were a history of ocular hypertension or glaucoma suspect, a history of glaucoma, the stage of diabetic retinopathy (i.e., none, non-proliferative, proliferative, or unspecified), and history of intravitreal anti-vascular endothelial growth factor (VEGF) injections, as anti-VEGF injections have been associated with increased glaucoma risk.[29,34,35] Systemic health state measures included histories of diabetes, hypertension, hyperlipidemia, and chronic kidney disease, body mass index (BMI), hemoglobin A1c (HbA1c) level, and the Diabetes Complications Severity Index (DCSI). The DCSI is a validated

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metric that is calculated using diagnosis codes from six categories of diabetic complications.[29] Other covariates included insulin, statin, beta-blocker, and calcium channel blocker usage within 45 days from the index date.

Statistical Analysis

The statistical analysis was conducted using the Python programming language (version 3.8.2, Python Software Foundation, Fredericksburg, Virginia, USA) and SQL queries on the UCHDW online Azure platform. When applicable, both eyes of each patient were included in the analysis. Descriptive analyses were conducted to evaluate the pre-exposure characteristics across the groups. The Shapiro-Wilk test was used to determine the normality of data distribution. Mann-Whitney U and Pearson's Chi-squared tests were used to compare the pre-exposure characteristics between the two groups for continuous and categorical variables, respectively. To compare mean pre- vs. post-exposure IOP, a paired T-test was conducted, including the eyes of each group. A violin graph was used to demonstrate the changes in IOP distribution in each group. Linear regression models were fitted and graphed on a scatter plot to assess the correlation between the ΔIOP and pre-exposure IOP. Further subgroup analysis in eyes with glaucoma and glaucoma suspects was conducted using a paired t-test. Similarly, eyes were grouped based on their pre-exposure IOP using the median as the cutoff, and subgroup analysis was conducted using GLP-1R agonist exposure and pre-exposure IOP as predictors of ΔIOP in Generalized Estimating Equations (GEE) analysis. Adjustment for pre-exposure IOP was performed to control for regression to the mean. Univariable regression analysis was performed using GEEs to evaluate the associations between covariates and the outcome measure. GEE is an accepted, widely used modeling approach in ophthalmology literature, which allows accounting for inter-eye correlations when including both eyes of the patients.[36,37] Variables with $p<0.1$ in the univariable GEE analysis were included in the multivariable GEE analysis. Sensitivity analyses were conducted to assess the robustness of the findings. P values <0.05 were considered statistically significant.

Results

Study Population

31584 tonometry records of 2330 eyes from 1173 individuals were included in the analysis. The overall cohort consisted of 1247 eyes of 626 individuals in the GLP-1R agonist group, with 1083 eyes of 547 patients in the other antidiabetics group.

Patient-Level Characteristics

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The median age of the included individuals was 66.2 years [IQR=18.3], with the greatest proportion being female (n=607 individuals, 51.7%) and Caucasian (n=667 individuals, 56.9%). Median pre-exposure IOP, HbA1c, and BMI were 15.2 mmHg [IQR=3.8], 7.5 [IQR=2.4], and 29.8 [IQR=9.4], respectively. 776 individuals (66.1%) had diabetes, with the median number of active oral antidiabetics being 1.0 [IQR=1.0], and 441 (37.5%) being insulin users. Several pre-exposure characteristics significantly differed between the GLP-1R Agonist and the control group, including age, gender, and race (Table 1).

Eye-Level Characteristics

A diagnosis of glaucoma had been made in 231 eyes (19.7%), and 269 eyes (22.9%) were diagnosed as glaucoma suspects. While the mean number of tonometry records was similar between groups (p=0.080), eyes in the GLP-1R Agonist group had more pre-exposure tonometry records (median = 2.0 [IQR = 2.0] vs. 1.0 [IQR = 1.0], p=0.002) and fewer post-exposure records compared to those in the control group (median = 1.0 [IQR = 2.0] vs. 2.0 [IQR = 2.0], p=0.002). Table 1 provides further information about the differences between the groups.

[Table 1 here]

Observed Changes in Intraocular Pressure

The mean and median time periods between medication exposure and first IOP measurement were 106.4 and 71.0 days, respectively. The mean ΔIOP was -0.4±2.8 mmHg (paired t-test, p<0.001) and -0.2±3.3 mmHg (paired t-test, p = 0.055) in the GLP-1R agonist and other antidiabetics groups, respectively (Figure 2). In subgroup analysis, significant decreases in IOP post-exposure were observed in eyes treated with semaglutide, liraglutide, and SGLT2 inhibitors (Table 2).

[Figure 2 here]

[Tables 2 here]

ΔIOP was significantly correlated with pre-exposure IOP in both the GLP-1R agonist and control groups (Figure 3). Eyes with a pre-exposure IOP ≥ 15.0 mmHg had an average ΔIOP of -1.44 mmHg (p<0.001), whereas eyes with a pre-exposure IOP <15.0 mmHg had an average ΔIOP of +0.94 mmHg (p<0.001). The supplemental figure shows the regression of ΔIOP against the pre-exposure IOP. In the group with pre-exposure IOP above 15.0, the decrease in IOP trended towards being greater for the GLP-1R agonist group than for the other antidiabetics group (coeff= -0.26, p=0.136) when adjusted for pre-exposure IOP. In the group with pre-exposure IOP below 15.0, the decrease in IOP trended towards being greater for the GLP-1R agonist group than for the other antidiabetics group (coeff= -

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0.02, p=0.879) when adjusted for pre-exposure IOP. Subgroup analysis of ΔIOP based on glaucoma diagnosis is summarized in Table 3.`

[Table 3 here]

Effect on Intraocular Pressure Variability

Filtering eyes with at least 3 pre-exposure and 3 post-exposure tonometry records resulted in a sample of 8738 tonometries performed on 327 eyes of 167 patients. Median pre-exposure IOP standard deviations in the GLP-1R agonist and control groups were 3.0 mmHg [IQR=6.0] and 3.0 mmHg [IQR=7.0], respectively. There were no significant differences in IOP standard deviation post-exposure, with GLP-1R agonist group having a post-exposure IOP standard deviation of 5.0 mmHg [IQR=5.5; p=0.197] and the control group with 5.1 mmHg [IQR=6.0; p=0.192]. Semaglutide (p=0.512), liraglutide (p=0.303), and SGLT2 inhibitors (p=0.237) were not significantly associated with changes in IOP variance post-exposure.

Generalized Estimating Equations

Although the paired analyses demonstrated that GLP-1R agonist exposure was associated with a significant decrease in IOP, in a multivariable regression model with GEEs accounting for inter-eye correlation and other covariates, GLP-1R agonists were not independently associated with ΔIOP (Table 4). Pre-exposure IOP and hypertension were independent predictors of ΔIOP following exposure to antidiabetics. These results were robust to the sensitivity analyses of different medication subgroups, yielding similar results.

[Table 4 here]

Discussion

There has been a growing interest in examining the effects of GLP-1R agonists for treating glaucoma. Neuroprotective effects of this class of medications have been shown in pre-clinical models of ocular diseases with a neurodegenerative component, including glaucoma.[38,39] GLP-1R agonists have been shown to mitigate neuroinflammatory processes and offer protection against glaucomatous neuronal damage.[31] Two large-scale retrospective case-control studies both reported a lower risk for a new glaucoma diagnosis in patients taking GLP-1R agonists compared to other antidiabetic medications.[29,30] Inspired by evidence of IOP lowering in a mouse model of hypertensive glaucoma exposed to GLP-1R agonists, this report is among the first to examine the effect of GLP-1R agonists on IOP. We reported the following main findings, to be discussed in turn: (1) a small, but

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statistically significant decrease in IOP following GLP-1R agonist initiation using paired t-test analysis, (2) an IOP decrease that is more prominent among eyes with pre-exposure IOPs ≥ 15 mmHg and among those with a history of glaucoma, and (3) GLP-1R agonist exposure was not an independent predictor of IOP in multivariable regression models when other covariates were included.

First, a significant decrease in IOP was noted in the GLP-1R agonist group following medication initiation, whereas a significant change was absent in those initiated on another antidiabetic agent. Among GLP-1R agonists, semaglutide and liraglutide, the most frequently initiated GLP-1R agonists in our cohort, were associated with a significant change in IOP. Interestingly, liraglutide and oral semaglutide were associated with the greatest decrease in IOP despite their comparatively smaller numbers, a finding at least in part consistent with previously reported preclinical data.[31] Injectable and oral formulations of semaglutide demonstrated similar levels of ΔIOP. It is worth noting, however, that the median IOP decrease following agonist exposure was 0.4 mmHg, and may not be clinically relevant. This limited effect size may be attributable to variabilities in the route and dosage of administration not captured by queries of the database. Although several pre-exposure characteristics differed between the two groups, it should not affect the results of the paired t-test analyses, as the paired t-tests involve analyzing the distribution of changes in IOP for each individual before and after the exposure, rather than comparing population-based means. Another class of medication that was associated with ΔIOP in this study was SGLT2 inhibitors. Chieh Shao et al. found that SGLT2 inhibitors reduced incident glaucoma over GLP-1R agonists in a large multi-institutional cohort of type 2 diabetic patients of Asian descent.[40] In comparison, while our study demonstrated that SGLT2 inhibitors slightly lowered IOP, ΔIOP was greater in the GLP-1R agonist group compared to the SGLT2 group.

Second, ΔIOP demonstrated a significant correlation with pre-exposure IOP, with the most notable change observed in eyes with a pre-exposure IOP of 15 mmHg or greater. Within this subgroup, exposure to GLP-1R agonists resulted in a mean ΔIOP of -1.44 mmHg, suggesting a potentially clinically significant IOP reduction in this subgroup (Table 3). This is consistent with preclinical data, which demonstrated IOP decrease in ocular hypertensive eyes but not in eyes without elevated IOP following GLP-1R agonist administration.[31] Further, ΔIOP was more prominent in glaucomatous eyes. In this study, we examined all individuals exposed to GLP-1R agonists to leverage the largest sample size possible for our statistical analyses. However, based upon these findings that showed a

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modest reduction among "all-comers" but a more substantial (and potentially clinically significant) reduction among certain subgroups, future analyses in specific subgroups such as ocular hypertensive and glaucomatous eyes are merited.

Last, when compared to controls consisting of exposure to other antidiabetic medications, GLP-1R agonists were not independent predictors of IOP reduction in multivariable GEE models that adjusted for additional covariates. However, several considerations should be noted when interpreting these results. First, the paired analysis showed that *within* the GLP1-R agonist group, there was a statistically significant reduction in IOP after exposure, but the multivariable GEE demonstrates that this difference was not statistically significant when comparing *across* groups (i.e., comparing the GLP1-R agonist group to the controls exposed to other antidiabetic medications). One possibility is that in the multivariable model, other factors were stronger drivers of IOP reduction. For example, our multivariable GEE suggests that pre-exposure IOP may be strongly associated with ΔIOP following exposure to antidiabetics. We investigated different cohorts and accounted for different confounding variables in the sensitivity analyses, all of which confirmed the primary results in the general population. There may also have been additional confounding factors that were not accounted for in our models, such as variations in prescription patterns, patient preferences, medication tolerability, and other factors that are not easily discerned from EHR data and outside the scope of this study. Finally, similar to our previous comments addressing subgroup analyses by pre-exposure IOP and glaucoma status, it is possible that multivariable GEEs within specific groups (i.e., those diagnosed with glaucoma) may yield different results. A known confounding factor when dealing with longitudinal IOP data is regression to the mean[41], a phenomenon that has been reported even in prospective landmark clinical trials such as the ocular hypertension treatment study (OHTS). [42] The observed trend in ΔIOP may have been in part due to regression to the mean. Aiming to account for this possibility, we ran a GEE analysis adjusting for pre-exposure IOP, showing that GLP-1R agonist exposure may be associated with a greater IOP decrease, a correlation that did not reach statistical significance but should be further explored in future studies.

These results broadly align with previous knowledge of GLP-1R agonists gained from preclinical studies.[31] A clue hinting at a possible mechanism for GLP-1R agonist-associated IOP reduction lies in its recent evaluation as a medical option for managing idiopathic intracranial hypertension.[43] Specifically, GLP-1R agonists have been shown

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to decrease cerebrospinal fluid production by inhibiting Na+/K+ ATPase activity within the choroidal plexus.[44] Aqueous humor secretion is similarly dependent upon Na+/K+ ATPase activity, and if GLP-1R agonists directly inhibit Na+/K+ ATPase within the ciliary body, this is one pathway through which aqueous secretion and, by extension, IOP might be decreased. Although weight loss, a known and desirable effect of GLP-1R agonist use, could play a role in IOP determination,[45] in our study, ΔBMI did not show any association with ΔIOP. However, with only 935 ΔBMI data points, this analysis was borderline powered to decisively comment on this association. Therefore, no causal analysis was pursued in this effort. Given the observed signals suggesting changes in IOP, along with biological plausibility and alignment with prior preclinical findings, further investigation is warranted to elaborate on the main findings of our study.

Results should be interpreted with the inherent limitations of retrospective studies in mind. We restricted the comparison group to individuals receiving antidiabetic medications to facilitate comparisons of ΔIOP by using a similar index date for both groups. An alternative approach could have been to compare those taking GLP-1R agonists to *all* individuals *not* taking GLP-1R agonists (i.e., include those who are not taking any antidiabetic medications), but this would have presented difficulty in determining corresponding index dates, which could have generated its own bias. Additionally, although no difference was found when comparing the distribution of tonometry recording times between groups, indicating similar measurement times, all IOP records were from daytime clinic visits, and may have missed diurnal variations and peak IOPs. Given these limitations, future prospective studies with standardized protocols are needed to further elucidate the effects of GLP-1R agonists on IOP.

Our study had multiple strengths. This was a retrospective cohort study using the UCHDW, a data repository that includes clinical data of 8.7 million patients across the University of California (UC) health network.[33] This data warehouse is unique for the purpose of this study in that it is one of the only big data resources that simultaneously includes data on systemic medication usage and IOP values. The Intelligent Research In Sight (IRIS) Registry [46] of the American Academy of Ophthalmology includes IOP values but does not have robust data regarding systemic (non-ocular) medications. Other well-known data repositories, such as the National Institutes of Health *All of Us* Research Program,[47] include systemic medications and clinical information but do not currently have a robust representation of IOP values.[48] This study is well-powered to evaluate the outcome measure with a sample size of

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2330 eyes of 1173 individuals. With an alpha of 0.05 and a beta of 0.8, a sample size of 101 eyes in each group (totaling 202 eyes) would be necessary to detect reductions of ≥1 mmHg. To detect a difference of 0.2 mmHg, the study would require 3771 eyes in each group (totaling 7542 eyes). Given the results of this power analysis, our study falls between these two extremes and is well-powered to state that the difference in effect sizes between the groups is less than 1 mmHg. Another strength of our present study was the statistical analysis, which accounted for potential confounding variables and allowed the generation of robust results. Furthermore, our study was guided by strong preclinical evidence. It stands as one of the initial clinical studies describing the association between GLP-1R agonists exposure and ΔIOP in humans and comparing it to other oral antidiabetics using real-world data. Future prospective studies with study protocol providing for dosage standardization may provide further insights.

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Table 1. Pre-exposure characteristics of eyes in both groups.

Variable	GLP-1R Agonist	Other Antidiabetics	p-value
Patient Level Characteristics	N=626	N=547	
Age (years)			**<0.001**[a]
Median [IQR]	64.0 [17.6]	67.8 [18.3]	
Gender, n (%)			
Female	351 (56.1)	256 (46.8)	**0.004**[b]
Male or unknown[c]	275 (43.9)	291 (53.2)	
Race, n (%)			0.019[b]
Caucasian	382 (61.0)	285 (52.1)	
Asian	109 (17.4)	124 (22.7)	
African American	16 (2.6)	15 (2.7)	
Other	119 (19.0)	123 (22.5)	
Diabetes, n (%)	433 (69.2)	343 (62.7)	**0.023**[b]
Number of active oral medications			0.342[a]
Median [IQR]	1 [1]	1 [1]	
Insulin, n (%)	253 (40.4)	188 (30.4)	**0.038**[b]
Statin, n (%)	451 (72.0)	394 (72.0)	1[b]
Diabetic Complications Severity Index			**0.029**[a]
Median [IQR]	6 [6]	5 [6]	
Hypertension, n (%)	376 (60.1)	318 (58.1)	0.541[b]
Hyperlipidemia, n (%)	377 (60.2)	276 (50.5)	**<0.001**[b]
Chronic Kidney Disease, n (%)	122 (19.5)	85 (15.5)	**0.090**[b]
Hemoglobin A1C			**<0.001** [a]
Median [IQR]	7.8[2.5]	7.3[2.3]	
Body mass index			**<0.001**[b]
Median [IQR]	31.8 [9.40]	27.4 [7.5]	
History of intravitreal anti-VEGF injection, n (%)	65 (10.4)	61 (11.2)	0.742[b]

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Diabetic retinopathy status, n (%)			0.393 [b]
No diabetic retinopathy	480 (76.7)	430 (78.6)	
Non-proliferative	80 (12.8)	54 (9.9)	
Proliferative	61 (9.7)	60 (11.0)	
Other or unspecified	5 (0.8)	6 (5.5)	
Ocular Hypertension, n (%)	77 (12.3)	59 (10.8)	0.474[b]
Glaucoma, n (%)	65 (10.4)	51 (9.3)	0.611[b]

GLP-1R: Glucagon-like protein-1 receptor, IQR: interquartile range, SD: standard deviation, IOP: intraocular pressure, a: mann-whitney u test, b: Chi-square test, c: The information for patients with unknown gender was collated with the male group due to the small sample size and regulations from the UC Health Data Warehouse, which prohibits reporting data for N<10 to prevent re-identification of patients.

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Table 2. Changes in intraocular pressure pre- and post-exposure to multiple antidiabetic medication groups.

Group	Number of eyes	Pre-exposure Median [IQR]	Post-exposure Median [IQR]	Mean Δ IOP	p-value[a]
All GLP1 receptor agonists	1247	15.3 [4.0]	15.0[4.0]	-0.4±2.8	<0.001
GLP1 receptor agonist subclasses[b]					
semaglutide	675	15.3 [4.0]	15.0 [4.0]	-0.5	<0.001
Injectable	517	15.3 [3.7]	15[4]	-0.5	<0.001
Oral	158	15.3[4.4]	15[4]	-0.6	0.016
dulaglutide	375	15.3 [4.0]	15.0 [3.9]	-0.3	0.06
liraglutide	137	16.0 [4.0]	15.0 [3.4]	-0.6	0.006
exenatide	50	15.3 [3.5]	15.5 [3.5]	0	0.923
All other antidiabetics	1083	15.0[4.0]	15.0 [4.0]	-0.2±3.3	0.055
Other antidiabetics subclasses					
Sodium-glucose co-transporter 2 (SGLT2) inhibitors	485	15.0 [4.0]	14.7 [5.0]	-0.4	0.005
Dipeptidyl peptidase 4 (DPP-4) inhibitors	273	15.0 [4.0]	15.0 [4.0]	0.1	0.756
Sulfonylureas	246	15.0 [4.0]	15.0 [4.0]	-0.1	0.513
Thiazolidinediones	79	16.0 [4.5]	15.0 [3.1]	-0.2	0.628

SGLT2: sodium-glucose co-transporter 2, DPP-4: dipeptidyl peptidase 4, SD: standard deviation, IQR: interquartile range, a: paired T test. b: An additional subclass, albiglutide, is not reported here due to the small sample size and regulations from the UC Health Data Warehouse, which prohibits reporting data for N<10 to prevent re-identification of patients.

Table 3. Comparison of pre- vs. post-exposure intraocular pressure (IOP) in glaucoma and glaucoma suspect eyes.

	GLP-1R Agonists					Other Antidiabetics				
Subgroup	**N eyes**	**Pre-exposure Median [IQR]**	**Post-exposure Median [IQR]**	**Mean ΔIOP**	**p[a]**	**N eyes**	**Pre-exposure Median [IQR]**	**Post-exposure Median [IQR]**	**Mean ΔIOP**	**p[a]**
Glaucoma	129	16.0 [4.3]	15.0 [2.9]	-0.7	0.002	102	15.4 [4.5]	15.0 [3.8]	-0.6	0.038
Glaucoma Suspect	86	16.6 [3.7]	16.0 [3.8]	-0.2	0.448	72	15.0 [4.9]	16.0 [4.6]	0.1	0.852
Others	1032	15.0 [3.7]	15.0 [4.0]	-0.4	<0.001	909	15.0 [4.0]	15.0 [4.0]	-0.2	0.119

IOP: intraocular pressure, a: paired t-test comparing pre- and post-exposure IOP of each eye.

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Table 4. Generalized Estimating Equation Analysis of Δ Intraocular Pressure (mmHg) Accounting for Inter-eye Correlation.

Variable	Univariable Analysis		Multivariable Analysis	
	Coefficient [95% CI]	p-value	Coefficient	p-value
Age	0.00 [-0.001 - 0.000]	0.354		
Gender	-0.004 [-0.025 - 0.017]	0.699		
Race (ref: Caucasian)				
African American	0.028 [-0.030 - 0.086]	0.341		
Asian	-0.019 [-0.046 - 0.008]	0.172		
Other	0.008 [-0.017 - 0.034]	0.53		
Pre-exposure IOP	**0.026 [0.019 - 0.032]**	***<0.001**	**0.026 [0.019 - 0.032]**	***<0.001**
Glaucoma diagnosis	**0.027 [-0.003 - 0.057]**	**0.074**	0.003 [-0.025 - 0.031]	0.841
Ocular hypertension	-0.003 [-0.035 - 0.029]	0.841		
Statin use	-0.018 [-0.042 - 0.006]	0.136		
Chronic kidney disease	-0.017 [-0.044 - 0.009]	0.193		
Hyperlipidemia	**0.025 [0.003 - 0.047]**	***0.023**	0.008 [-0.015 - 0.031]	0.499
Diabetes	0.003 [-0.020 - 0.025]	0.8		
Diabetic retinopathy severity	-0.000 [-0.014 - 0.013]	0.967		
Hypertension	**0.026 [0.004 - 0.048]**	***0.020**	0.024 [-0.000 - 0.047]	0.053
Intravitreal Anti VEGF injection	0.008 [-0.019 - 0.034]	0.557		
DCSI Score	0.000 [-0.002 - 0.003]	0.789		
Oral antidiabetics count	-0.011 [-0.025 - 0.003]	0.116		
Insulin use	-0.007 [-0.028 - 0.015]	0.552		
Beta blockers	-0.012 [-0.033 - 0.010]	0.291		
Calcium channel blockers	0.006 [-0.016 - 0.027]	0.611		
ΔBMI	-0.007 [-0.018 - 0.004]	0.202		
ΔHbA1c	**-0.017 [-0.038 - 0.004]**	**0.115**	-0.013 [-0.032 - 0.007]	0.202
GLP-1R agonist exposure	0.000	0.946		

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	[-0.005 - 0.006]

IOP: intraocular pressure, VEGF: vascular endothelial growth factor, DCSI: diabetes complications severity index, BMI: body mass index, HbA1c: hemoglobin A1c, GLP-1R: glucagon-like peptide 1 receptor.

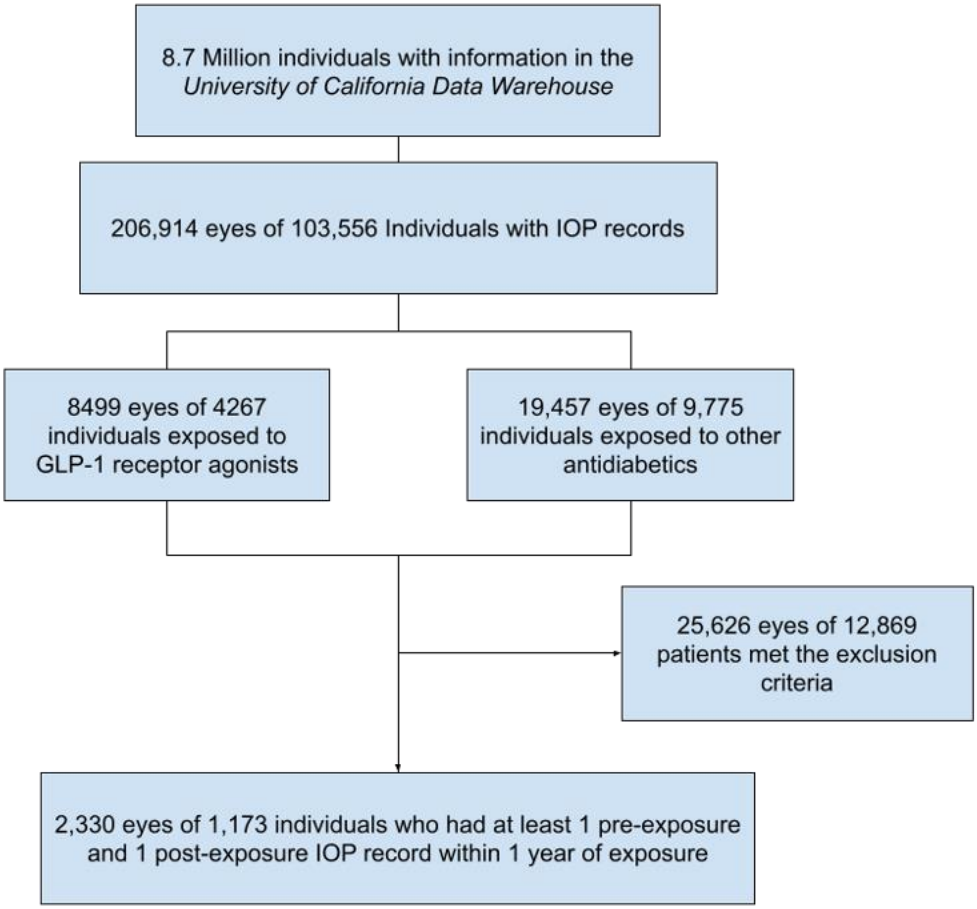

Figure 1. Flowchart depicting study cohort selection process. IOP: intraocular pressure, GLP-1R: glucagon-like peptide 1 receptor.

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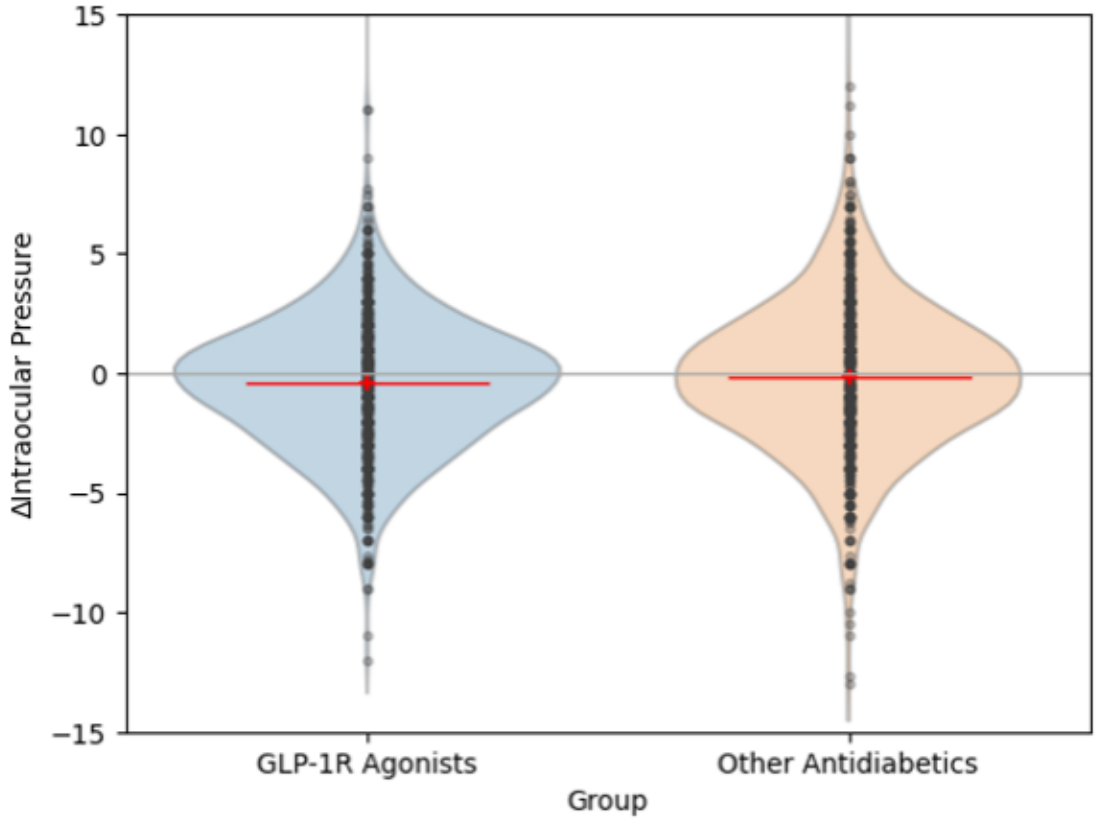

Figure 2. Violin plots comparing the distribution of Δ intraocular pressures (IOPs) between the groups. This change was only significant in the paired analysis of the individuals in the glucagon-like peptide 1 receptor (GLP-1R) agonists group.

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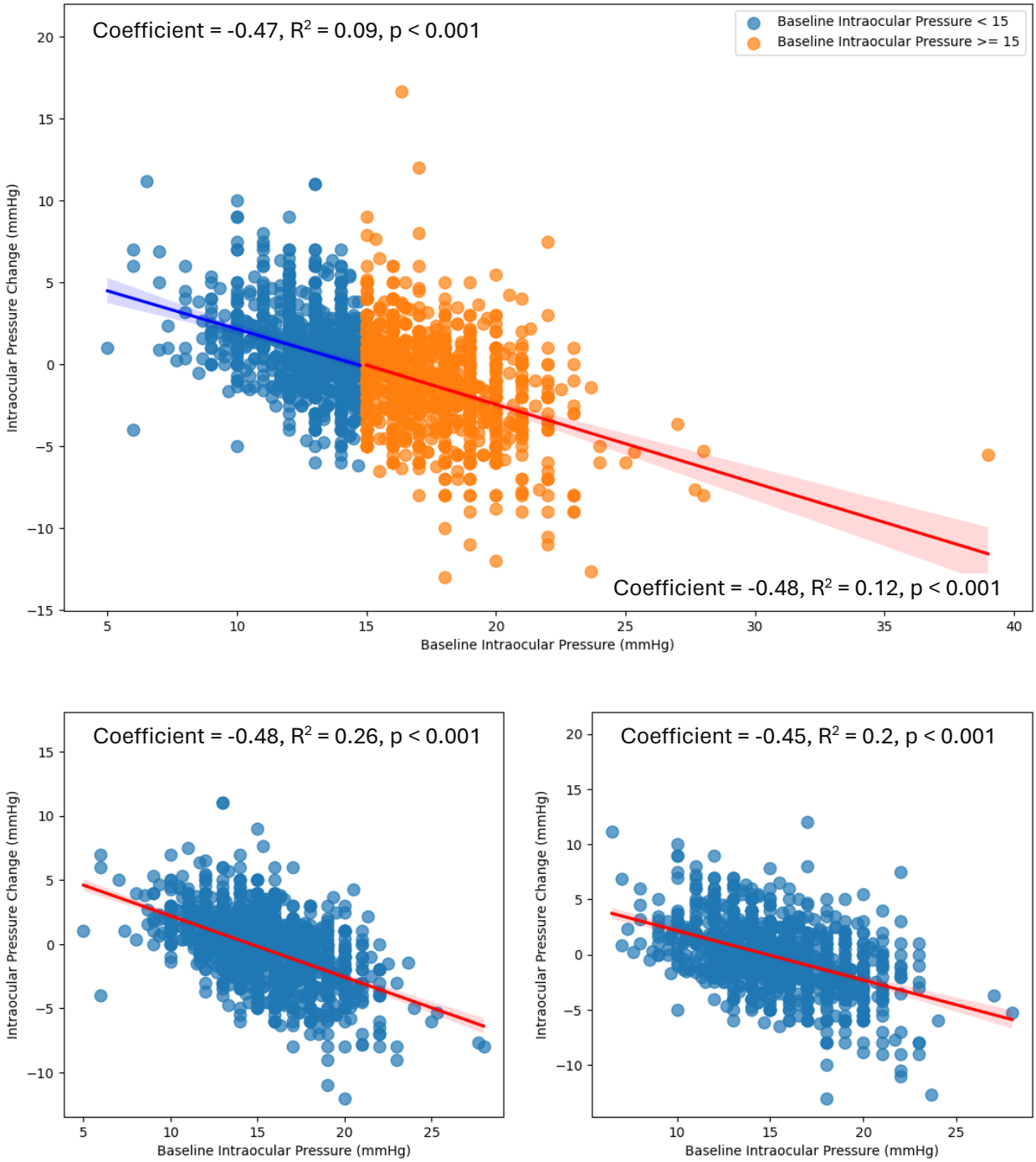

Supplemental Figure. Scatterplot of intraocular pressure changes over pre-exposure intraocular pressures in the entire study population *(top)*, in the glucagon-like peptide 1 receptor agonist group (*bottom left)*, and the control group (*bottom right*).

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Supplementary Table 1

VARIABLE	OMOP CONCEPT NAME	CONCEPT ID	VOCABULARY
INTRAOCULAR PRESSURE			
	Intraocular pressure of left eye	44805433	SNOMED
	Intraocular pressure of right eye	44813337	SNOMED
GLP-1R			
	liraglutide; parenteral	1123630	ATC
	exenatide; parenteral	1123633	ATC
	albiglutide; parenteral	1123739	ATC
	semaglutide; parenteral	1501778	ATC
	lixisenatide; parenteral	1123609	ATC
	dulaglutide; parenteral	1123611	ATC
	beinaglutide; systemic	947847	ATC
CONTROL ANTIDIABETIC AGENTS			
	Dipeptidyl peptidase 4 (DPP-4) inhibitors	21600783	ATC
	Sodium-glucose co-transporter 2 (SGLT2) inhibi...	1123627	ATC
	Sulfonamides (heterocyclic)	21600763	ATC
	Sulfonylureas	21600749	ATC
	Thiazolidinediones	21600779	ATC
OTHER NON-INSULIN ANTIDIABETICS			
	metformin; oral	21600747	ATC
	Alpha glucosidase inhibitors	21600775	ATC

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	Other blood glucose lowering drugs, excl. insu...	21600788	ATC
	Combinations of oral blood glucose lowering drugs	21600765	ATC
ANTIGLAUCOMA MEDICATIONS			
	ANTIGLAUCOMA PREPARATIONS AND MIOTICS	21603671	ATC
GLAUCOMA PROCEDURE/SURGERY			
	Insertion of anterior segment aqueous drainage...	44816348	CPT4
	Insertion of anterior segment aqueous drainage...	2101785	CPT4
	Goniotomy	2110960	CPT4
	Trabeculotomy ab externo	2110961	CPT4
	Trabeculoplasty by laser surgery	2110962	CPT4
	Fistulization of sclera for glaucoma; sclerect...	2110977	CPT4
	Fistulization of sclera for glaucoma; trabecul...	2110979	CPT4
	Fistulization of sclera for glaucoma; trabecul...	2110980	CPT4
	Aqueous shunt to extraocular equatorial plate ...	2110981	CPT4
	Revision of aqueous shunt to extraocular equat...	2110982	CPT4
	Ciliary body destruction; cyclophotocoagulatio...	2110996	CPT4
	Ciliary body destruction; cyclophotocoagulatio...	2110997	CPT4
	Insertion of aqueous drainage device, without ...	42628005	CPT4
	Aqueous shunt to extraocular equatorial plate ...	46257451	CPT4
	Revision of aqueous shunt to extraocular equat...	46257632	CPT4
	Transluminal dilation of aqueous outflow canal...	40757003	CPT4
	Transluminal dilation of aqueous outflow canal...	40757013	CPT4
OCULAR HYPERTENSION OR GLAUCOMA SUSPECT			

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	Ocular hypertension	381290	SNOMED
	Glaucoma suspect	3663197	SNOMED
GLAUCOMA			
	Glaucoma	437541	SNOMED
DIABETIC RETINOPATHY STATUS			
	Retinopathy due to diabetes mellitus	4174977	SNOMED
	Nonproliferative retinopathy due to diabetes m...	376683	SNOMED
	Proliferative retinopathy due to diabetes mell...	380096	SNOMED
	Nonproliferative retinopathy of right eye due ...	37311253	SNOMED
	Nonproliferative retinopathy of left eye due t...	37311254	SNOMED
	Proliferative retinopathy of right eye with di...	36674199	SNOMED
	Proliferative retinopathy of left eye due to d...	36674200	SNOMED
ANTI-VEGF INJECTIONS			
	ranibizumab Injection [Lucentis]	35603582	RxNorm
	Avastin Injectable Product	36229370	RxNorm
	aflibercept	40244266	RxNorm
	Intravitreal injection of anti-vascular endoth...	604986	SNOMED
HISTORY OF DIABETES			
	Diabetes mellitus	201820	SNOMED
	Type 1 diabetes mellitus	201254	SNOMED
	Type 2 diabetes mellitus	201826	SNOMED
HISTORY OF HYPERTENSION			
	Hypertensive disorder	316866	SNOMED
HISTORY OF HYPERLIPIDEMIA			
	Hyperlipidemia	432867	SNOMED

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HISTORY OF CKD			
	Chronic kidney disease	46271022	SNOMED
BODY MASS INDEX (BMI)			
	Body mass index (BMI) [Ratio]	3038553	LOINC
HBA1C			
	Hemoglobin A1c [Mass/volume] in Blood	3034639	LOINC
	Hemoglobin A1c measurement	4184637	SNOMED
	HbA1c level (Diabetes Control and Complication...	37393623	SNOMED
	Hemoglobin; glycosylated (A1C) by device clear...	2212393	CPT4
	Hemoglobin A1c/Hemoglobin.total in Blood	3004410	LOINC
	Hemoglobin A1c/Hemoglobin.total in Blood by HPLC	3005673	LOINC
INSULIN			
	INSULINS AND ANALOGUES	21600713	ATC
BETA BLOCKER			
	BETA BLOCKING AGENTS	21601665	ATC
CALCIUM CHANNEL BLOCKERS			
	CALCIUM CHANNEL BLOCKERS	21601744	ATC
STATINS			
	HMG CoA reductase inhibitors	21601855	ATC

Supplementary Table 1: Concept IDs used for identifying records of the variables mentioned in this study. When extracting data, the database was queried for records with concept_ids equal to these codes, or to any descendant in the hierarchy of the OMOP vocabulary. DCSI codes are excluded for clarity, but the codes used are documented in the following repository: https://github.com/OHDSI/FeatureExtraction/blob/bb0213e64198b98e8fe80c3e964d308952779515/inst/sql/sql_server/Dcsi.sql. Acronyms: ATC = Anatomical Therapeutic Chemical; LOINC = Logical Observation Identifiers Names and Codes; CPT4 = Current Procedural Terminology 4; SNOMED = Systematized Nomenclature of Medicine Clinical Terms. RxNorm provides normalized names for clinical drugs.